

Luong attention

between two sentences

1. scores

$$e_{ij} = \textcolor{blue}{s}_i \cdot \textcolor{brown}{h}_j$$

2. weights

$$\alpha_{ij} = \text{softmax}_j(e_{ij})$$

3. output

$$c_i = \sum_j \alpha_{ij} \textcolor{green}{h}_j$$

Self-attention

inside one sentence

$$s_{ij} = \textcolor{blue}{q}_i \cdot \textcolor{brown}{k}_j$$

$$w_{ij} = \text{softmax}_j(s_{ij})$$

$$y_i = \sum_j w_{ij} \textcolor{green}{v}_j$$

query: decoder state $s_i \rightarrow$ query q_i of a word **key:** encoder state $h_j \rightarrow$ key k_j **value:** encoder state $h_j \rightarrow$ value v_j

same three steps; only where the query, key and value vectors come from changes